

E-Commerce Sales Analysis Project

E-COMMERCE SALES ANALYSIS USING SQL

SQL Database Project

Submitted by:

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Course: **Data Analytics** | Batch: **13-DA** | Institute Name :**Accio-Job**

Subject: **SQL / Database Management System**

Technology Used: **MySQL**

Introduction

E-commerce has become an important part of modern business. E-commerce platforms generate large amounts of data related to customers, products, orders, payments and sales. SQL can be used to store, manage and analyse this data efficiently.

This project focuses on analysing an e-commerce database using MySQL. The project aims to extract useful business insights such as sales performance, customer purchasing behaviour, product performance and payment trends using SQL queries.



Project Objectives

- **2. Objectives of the Project**
- The main objectives of this project are:
- To design an e-commerce database using MySQL.
- To store product, order and payment information.
- To analyse total sales and revenue.
- To identify top-selling products.
- To identify the most profitable product categories.
- To analyse payment methods and order status.
- To use SQL queries for extracting meaningful business insights.



Database Design

Database Structure

Orders table:

	order_id	customer_id	order_date
▶	1001	1	2024-01-05
	1002	2	2024-01-08
	1003	1	2024-01-15
	1004	3	2024-01-20
	1005	4	2024-02-02

orders 1 ×

Product_Orders table:

	product_id	order_id	quantity	unit_price
▶	1	3	5	72.69
	1	25	4	63.59
	1	59	3	22.07
	1	68	4	196.11
	1	71	5	227.35

product_orders 2 ×

Products table:

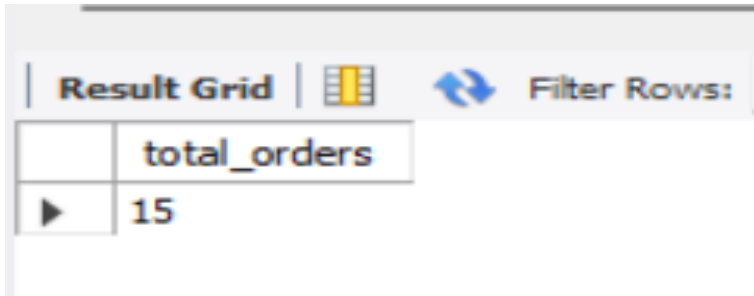
	product_id	product_name	category	brand	price
▶	P101	Laptop	Electronics	Dell	65000
	P102	Mobile	Electronics	Samsung	35000
	P103	Headphones	Electronics	Sony	5000
	P104	Office Chair	Furniture	IKEA	12000
	P105	Desk	Furniture	IKEA	18000

products 3 ×

SQL Queries / Analysis

Question 1 Total number of orders

```
SELECT COUNT(*) AS total_orders  
FROM orders;
```

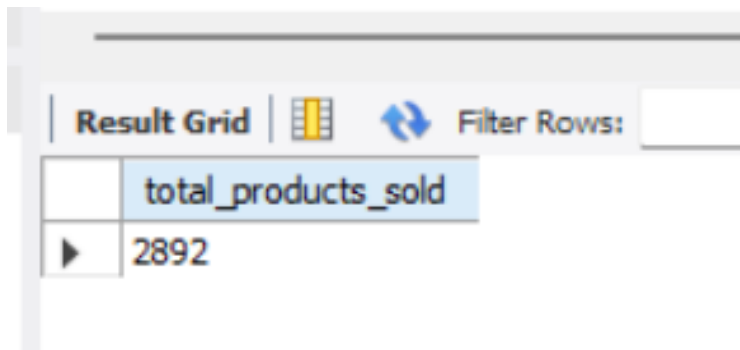


The screenshot shows a database interface with a toolbar at the top containing 'Result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' dropdown. Below the toolbar is a table with one row. The first column is empty, and the second column contains the text 'total_orders'. Below this, a second row shows a right-pointing triangle icon followed by the value '15'.

	total_orders
▶	15

Question 2 Total quantity of products sold

```
SELECT SUM(quantity) AS total_products_sold  
FROM product_orders;
```



The screenshot shows a database interface with a toolbar at the top containing 'Result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' dropdown. Below the toolbar is a table with one row. The first column is empty, and the second column contains the text 'total_products_sold'. Below this, a second row shows a right-pointing triangle icon followed by the value '2892'.

	total_products_sold
▶	2892

Question 3 Total revenue

```
SELECT round(SUM(quantity * unit_price),2) AS total_revenue
FROM product_orders;
```

Result Grid			Filter Rows:	
	total_revenue			
▶	465299.45			

Question 4 Average selling price

```
SELECT round(AVG(unit_price),2)AS average_selling_price
FROM product_orders;
```

Result Grid			Filter Rows:	
	average_selling_price			
▶	162.88			

Question 5 Highest priced product

```
SELECT product_id, product_name, category, brand, price
FROM products
ORDER BY price DESC
LIMIT 1;
```

Result Grid			Filter Rows:		Export:	
	product_id	product_name	category	brand	price	
▶	P101	Laptop	Electronics	Dell	65000	

Question 6 Lowest priced product

```
SELECT product_id, product_name, category, brand, price
FROM products
ORDER BY price ASC
LIMIT 1;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	product_id	product_name	category	brand	price
	P107	T-Shirt	Fashion	Adidas	1500

Question 7 Revenue generated by each product

```
SELECT
    p.product_id,
    p.product_name,
    SUM(po.quantity * po.unit_price) AS revenue
FROM products p
JOIN product_orders po
    ON CAST(SUBSTRING(p.product_id, 2) AS UNSIGNED) - 100 = po.product_id
GROUP BY p.product_id, p.product_name
ORDER BY revenue DESC;
```

Result Grid	Filter Rows:	Export:
product_id	product_name	revenue
P103	Headphones	9542.97
P106	Shoes	7541.13
P110	Printer	7323.43
P102	Mobile	7061.699999999999
P104	Office Chair	6049.710000000001

Result 17 x

Question 8 Rank products by revenue

```
SELECT
    product_name,
    revenue,
    DENSE_RANK() OVER (ORDER BY revenue DESC) AS revenue_rank
FROM (
    SELECT
        p.product_name,
        SUM(po.quantity * po.unit_price) AS revenue
    FROM products p
    JOIN product_orders po
    ON p.product_id = CONCAT('P', po.product_id + 100)
    GROUP BY p.product_id, p.product_name
) AS product_sales
ORDER BY revenue_rank;
```

Result Grid | | Filter Rows: | Export: | Wr

	product_name	revenue	revenue_rank
▶	Headphones	9542.97	1
	Shoes	7541.13	2
	Printer	7323.43	3
	Mobile	7061.699999999999	4
	Office Chair	6049.710000000001	5

Result 21 ▾

Question 9 Average order value

```
SELECT
    round(AVG(order_total),2) AS average_order_value
FROM (
    SELECT
        order_id,
        SUM(quantity * unit_price) AS order_total
    FROM product_orders
    GROUP BY order_id
) AS order_summary;
```

Result Grid | | Filter Rows:

	average_order_value
▶	1632.63

Question 10 Products whose sold quantity is more than 5

```
SELECT
    p.product_name,
    SUM(po.quantity) AS total_sold
FROM products p
JOIN product_orders po
    ON p.product_id = CONCAT('P', po.product_id + 100)
GROUP BY p.product_id, p.product_name
HAVING SUM(po.quantity) > 5
ORDER BY total_sold DESC;
```

Result Grid			Filter Rows:
	product_name	total_sold	
▶	Headphones	45	
	Mobile	43	
	Shoes	41	
	Laptop	37	
	Backpack	36	

Question 11 Products that were never ordered

```
SELECT
    p.product_id,
    p.product_name,
    p.category
FROM products p
left JOIN product_orders po
    ON p.product_id = po.product_id
WHERE po.product_id IS NULL;
```

Result Grid				Filter Rows:
	product_id	product_name	category	
▶	P101	Laptop	Electronics	
	P102	Mobile	Electronics	
	P103	Headphones	Electronics	
	P104	Office Chair	Furniture	
	P105	Desk	Furniture	

Result 26 ×

Key Findings / Insights

Key Findings

- Total revenue generated by the e-commerce business.
- Highest revenue-generating product.
- Best-selling product by quantity.
- Highest-performing category.
- Average order value.
- Products performing above average revenue.

Business Insights

The analysis identifies the products and categories contributing the most revenue.

High-performing products can be given greater promotional focus, while low-performing products can be reviewed for pricing or inventory decisions.

Sales trends also help understand changes in revenue over time.

Conclusion

This project demonstrates how SQL can be used to analyse e-commerce transactional data.

The analysis covers product performance, revenue, order trends and category-level sales.

SQL queries helped transform raw data into meaningful business insights.

The image features a soft, pink watercolor wash in the center, with a delicate, star-like pattern of small dots scattered across it. The words "Thank You" are written in a black, elegant cursive script, positioned centrally over the pink area. The overall aesthetic is gentle and appreciative.

Thank
You